

Homework 5

1. Read (at least) the abstract (first page) for the Green Space and Bone Density paper that is attached in Blackboard.

Please type your answers to the following questions into this document, and SAVE AS A PDF before uploading. Write in complete sentences, offering explanations for your answers. Avoid one-word answers.

- a. What are the cases in this study?

The cases are 327 individual children aged 4 to 6 years old who are a part of the environment birth study.

- b. **What is the main research question for this study?**

The main research question is whether early life exposure to surrounding green space is associated with changes in bone material density in young children.

- c. **What is the primary explanatory variable?**

The primary explanatory variable is the amount of residential green space surrounding the child's home, which was measured in different categories (high green, low green, and total green) within radii from 100 to 3000 meters.

- d. **Were any other explanatory variables recorded in this study? If so, tell what they are.**

Yes, other explanatory variables recorded include the different radii of green space measurements (100m to 3000m) and the classification of green space into high and low vegetation.

- e. **What is the main response variable?**

The main response variable is the children's bone mineral density, which was assessed using quantitative ultrasound measurement and recorded as the mean of the axially transmitted speed of sound in meters per second.

- f. **Were any other response variables recorded in this study? If so, tell what they are.**

Yes, another response variable recorded was the likelihood of a child having a bone density lower than the sex specific 10th percentile, which was analyzed as a risk factor in the study.

- g. **This study was observational. Explain how I know this.**

This study is observational because it did not manipulate the exposure to green space but instead measured naturally occurring differences in children's environments.

- h. If instead we wanted to conduct a randomized experiment to study the same research question, describe in a few sentences what that set-up would look like, even if you decide it would not be practical or ethical. (You can comment on that also.)**

A randomized experiment would involve assigning children or families to live in areas with varying amounts of green space for a set period and then measuring their bone mineral density over time. This setup would ensure that differences in bone health are due to green space exposure rather than other factors. Though, this would be impractical (and unethical too) because it would require relocating families.

- i. Can we draw causal conclusions from the study as it was conducted?**

No, we cannot draw causal conclusions because this is an observational study, and there might be unmeasured confounding variables that influence both the green space exposure and bone mineral density. This study suggests that there is an association it does not establish that green space directly causes improved bone health.

- j. Can we generalize the study to a larger population?**

Yes, they can be generalized to a similar population (assuming it's young children) living in urban or Suburban areas. Although, it is important to note that this may not be generalizable to individuals in different geographical locations and socio-economic backgrounds as outside factors such as nutrition and physical activity may vary across populations and regions.

2. **Go to [StatKey](#)**, select One Quantitative Variable, and in the dropdown menu that starts with “Mammal Longevity,” select “US 4-Year Colleges (Cost).” You should see a dotplot, as well as the summary statistics, for the total cost of attendance (including room and board) for 2,012 colleges that primarily grant bachelor’s degrees. These numbers all come from the Department of Education’s College Scorecard in 2019.

Write a description of that variable. Make sure to write in context, giving your “who/what” and also describing shape, center, and spread with appropriate numbers drawn from the graph and/or summary statistics table.

This graph represents the total annual cost of attendance at about 2,000 US four-year colleges. The distribution is right-skewed (it tails off toward higher costs) with a peak around the low-to-mid \$20,000s. The mean cost is roughly \$34,300, while the median is about \$30,700 according to the summary statistics table.